

Directional structure and environment

Version: 14/09/2026

Chapter 3, Sections 3.5.1–3.5.3

Thesis §3.5.1

East–north balance in the retained archive

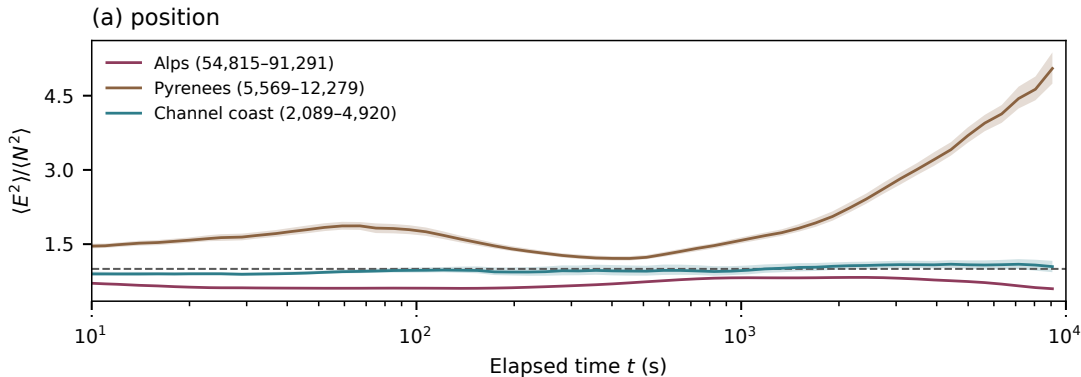
Equal east and north moments do not establish isotropy

$$\rho_{EN} = 1, \quad \Sigma = \begin{pmatrix} 1 & 0.9 \\ 0.9 & 1 \end{pmatrix}, \quad \lambda_1/\lambda_2 = 19.$$

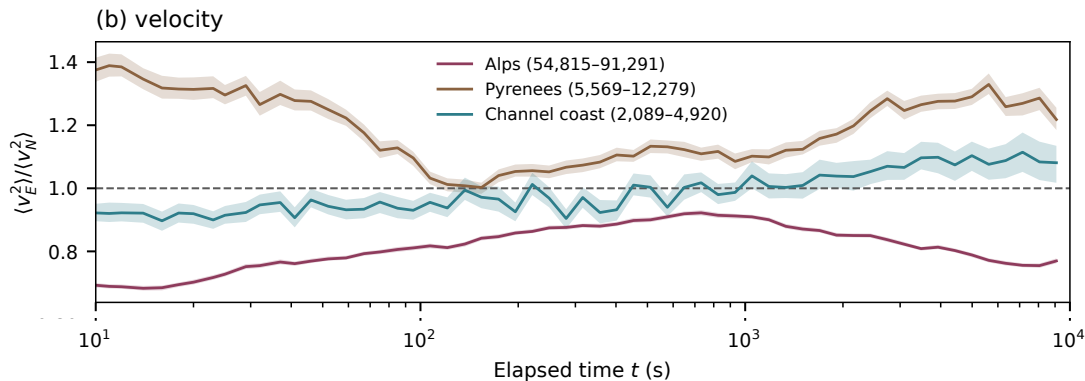
This centred Gaussian has balanced coordinate moments but strongly unequal principal variances.

A pooled mixture of east-directed and north-directed routes can also balance the ensemble while each flight remains directional.

Regional position moments differ along east and north



Regional velocity moments show a related directional contrast



Regional uncertainty bands do not match the groups

The ratios use paired east and north observations. A site–day bootstrap with 200 draws gives pointwise 10–90% ranges; each plotted time needs at least 30 flights and 10 clusters.

Missing site/date keys are treated as singletons. Resampling these groups does not match the equipment populations on weather, task or duration.

Thesis §3.5.2

Regional principal axes

PCA describes preferred axes; wind needs independent information

A rotation preserves distance

$$\Sigma = U\Lambda U^{\top}, \quad |U^{\top}\mathbf{X}| = |\mathbf{X}|.$$

PCA decorrelates components at a chosen lag. It cannot repair radial collapse or remove temporal dependence.

Whitening changes lengths and only fixes second moments at the fitted scale.

The following panels report regional PCA and its support. Anisotropy alone is compatible with a self-similar vector and a self-similar radius.

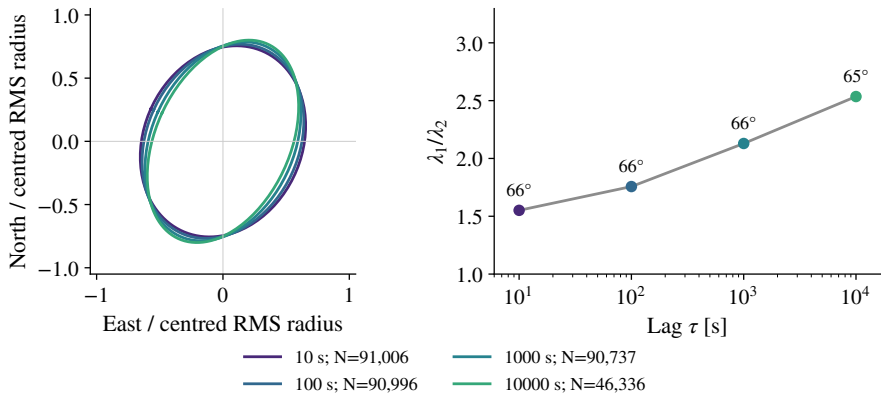
Ground velocity has two unknown terms

$$\mathbf{v}_{\text{ground}} = \mathbf{v}_{\text{air}} + \mathbf{w}_h.$$

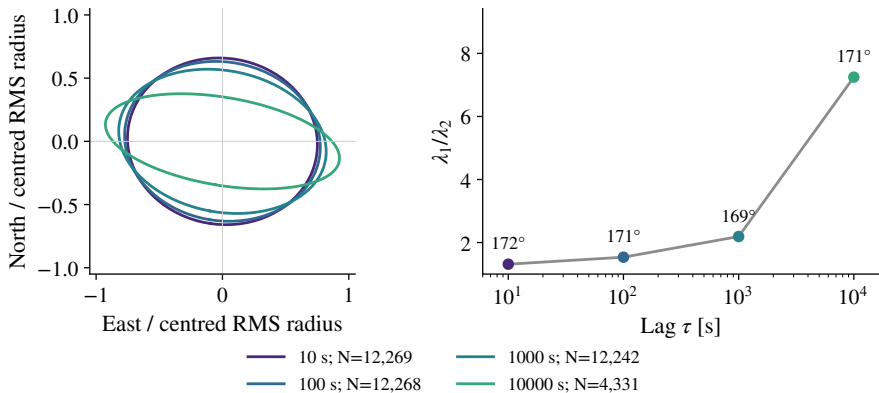
Terrain constrains headings; wind also changes position. Differentiation does not separate the two.

Different marginal exponents, or changing joint dependence, can prevent radial scaling.

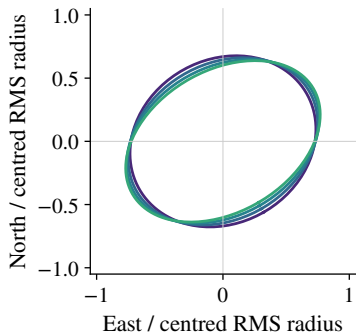
Alps: centred principal axes across four lags



Pyrenees: centred principal axes across four lags

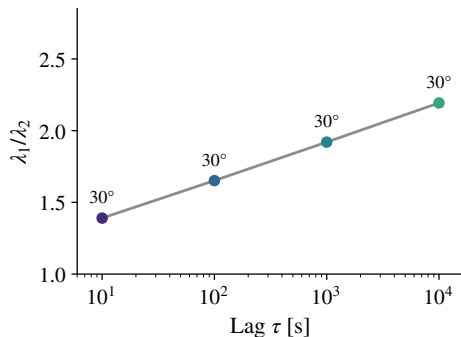


Channel Coast: centred principal axes across four lags



10 s; N=4,914

100 s; N=4,913



1000 s; N=4,890

10000 s; N=1,680

A terrain direction measured independently of the flights

NOAA ETOPO 2022

Same Alpine and Pyrenean take-off boxes.

One-arc-minute elevation sampling; retain the footprint above 1000 m.

Project cell centres to regional east/north coordinates. The area-weighted spatial covariance gives the main axis.

[NOAA NCEI, ETOPO 2022](#). Both elevation subsets, requests, code and hashes are archived. Angles are counterclockwise from east, modulo 180° .

Definition sensitivity

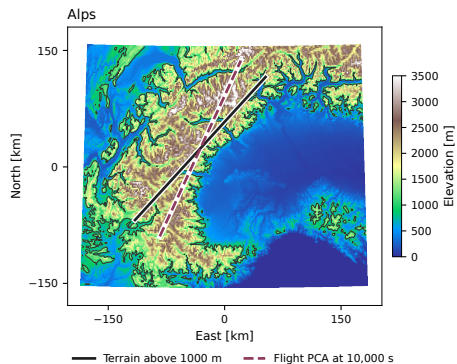
Thresholds: 500, 1000, 1500 m.

Alps: $42.1\text{--}47.9^\circ$.

Pyrenees: $169.7\text{--}175.4^\circ$.

These spans are not confidence intervals; the axis describes an area, not a traced crest.

Alps: shared sector, with an approximately 18-degree offset



Terrain: 47.6° .

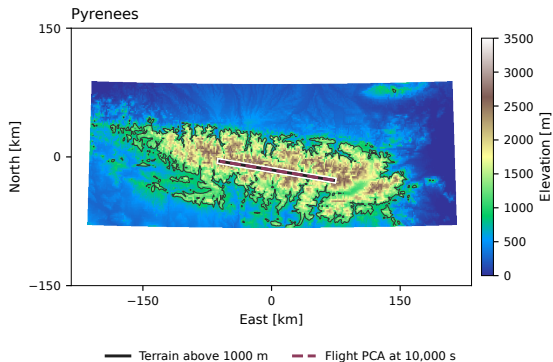
Flight PCA: 65.2° .

Separation: 17.6° .

46,336 flights support 10,000 s.

Black: highland-footprint axis. Dashed colour: current flight PCA, shifted to the same centre. Line lengths have no physical meaning.

Pyrenees: the long-lag spread follows the highland axis



Terrain: 170.5° .

Flight PCA: 170.5° .

Separation: less than 1° .

4,331 flights support 10,000 s.

ETOPO footprint above 1000 m; threshold sensitivity: $169.7\text{--}175.4^\circ$. Decimal agreement does not imply this physical precision.

Same population as the 10,000-s PCA

- 1,680 paraglider flights.
- 2,013 nonoverlapping windows, wholly inside retained segments.
- Mean cleaned GNSS altitude along each window; equal window weights.

Mean: 1,006 m

Equal flight weights: 980 m.

Middle half of window means:
854 m–1,160 m.

Approximate altitude above sea level;
recorder datums are not harmonised.

Trapezoidal integration on the same 10-s segment grids. The target describes the supported long-lag windows; it is not take-off altitude or height above the terrain.

ERA5 wind is interpolated to the measured flight altitude

2016–2025, hourly; nine cells in the coastal box; 1000/925/850/700 hPa.

$$H = \Phi / g_0, \quad h = \frac{R_{\oplus} H}{R_{\oplus} - H}, \quad \theta_w = \text{atan2}(\langle v(h) \rangle, \langle u(h) \rangle).$$

Convert geopotential to each hour's level heights. Interpolate **E/N wind components** to 1,006 m, between above-ground bounds.

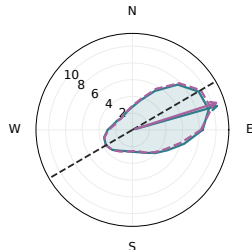
Main comparison: all year, all hours vs. all year, 09–17 UTC.

April–September is a separate seasonal check. Winds at 10/100 m AGL provide lower-level controls.

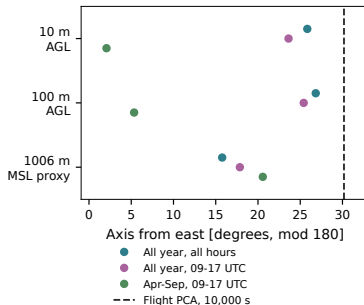
[ECMWF/Copernicus ERA5 pressure levels](#), accessed through a pinned snapshot of [Earthmover's public cloud-optimised ERA5 copy on AWS](#). Earthmover distributes this lossless copy; it is not a separate reanalysis. Complete hourly extract: 32.22 MB compressed, preserved in the repository and on SSD.

At flight height, both annual references remain offset from PCA

At mean flight altitude: 1006 m
Towards-sector frequency [%]



Height and time selection



Annual gaps at flight height: 14.4° for all hours, 12.3° for 09–17 UTC.

Angles counterclockwise from east. The rose shows where air moves towards at 1,006 m; the black line is the 30.2° centred PCA axis.

Height and time checks preserve a broad orientation resemblance

At 1,006 m	Wind axis	PCA gap	ρ_w
All year, all hours	15.8°	14.4°	0.43
All year, 09–17 UTC	17.9°	12.3°	0.44
Apr–Sep, all hours	16.0°	14.2°	0.35
Apr–Sep, 09–17 UTC	20.6°	9.6°	0.37

At the quartile heights, annual axes span 14.3–17.2° (all hours) and 16.4–19.3° (daytime), on shared support.

$\rho_w = \|\langle \mathbf{w} \rangle\| / \langle \|\mathbf{w}\| \rangle$ measures cancellation. Height ranges are sensitivities, not confidence intervals. The main estimates retain all selected hours.

What the environmental comparisons establish

Region	Conclusion supported by this comparison
Pyrenees	Close terrain–flight alignment; a geographical association worth testing along tracks.
Alps	Same broad sector, with an approximately 18-degree offset from the mean highland axis.
Channel Coast	Wind at mean flight altitude lies about 12–14 degrees from PCA in both annual time selections; seasonal checks retain a broad resemblance.

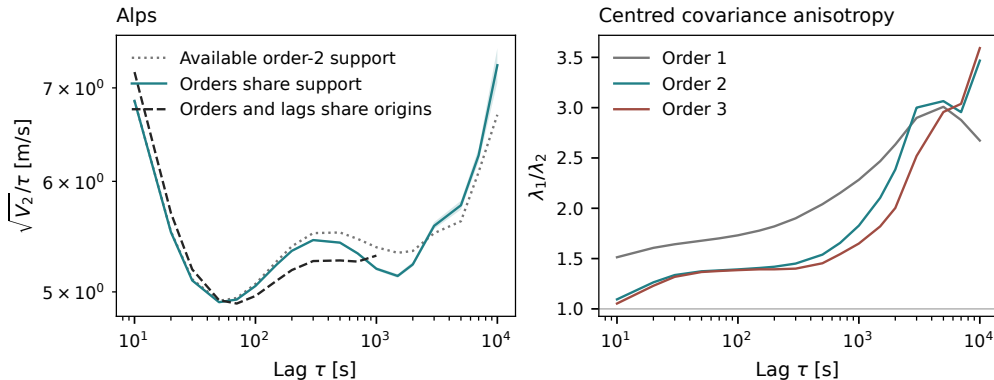
A common constant wind shifts the mean displacement by $\tau\mathbf{w}$; centring removes it. Variable winds and route responses can still shape covariance.

Independent regional references constrain an interpretation. They do not separate wind, terrain, route choice and long-flight selection.

Thesis §3.5.2

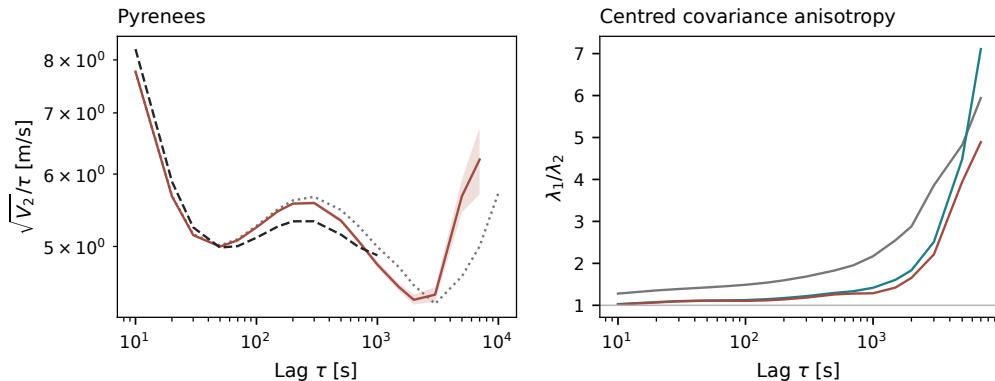
Regional changes of velocity and direction

Alps: velocity changes retain a directional structure



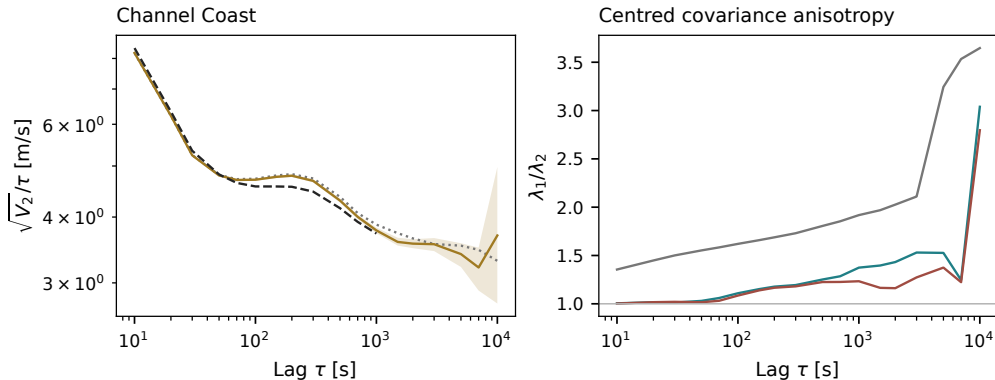
Equal flight weights. At 1000 s, matched orders retain 89,181 flights. Shading: pointwise 95% site-day intervals.

Pyrenees: the long-lag change coincides with falling support



At 1000 s: 12,020 matched flights; at 10,000 s: only 3. Cells below eight flights are omitted from plots.

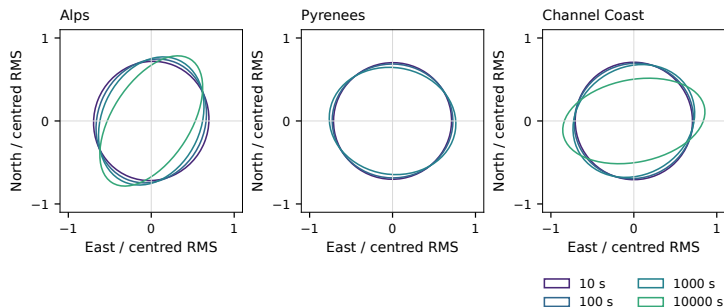
Channel Coast: smaller changes at 1000 s, with residual anisotropy



At 1000 s: 4,679 matched flights, $D_2 = 3.77$ m/s and covariance ratio 1.37.

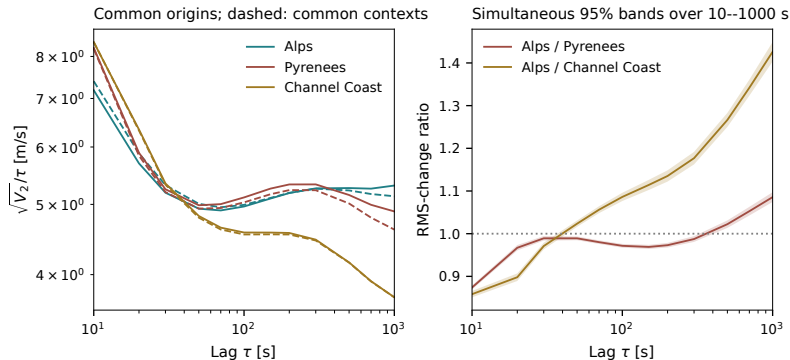
Second-difference ellipses retain orientation after drift cancellation

$$C_2 = \text{Cov}(A_2 \mathbf{r}), \quad V_2 = \text{tr } C_2 + |\langle A_2 \mathbf{r} \rangle|^2.$$



Matched-order origins, equal flight weights; each ellipse is normalised by its own centred RMS.

The regional contrast persists in a common-context control

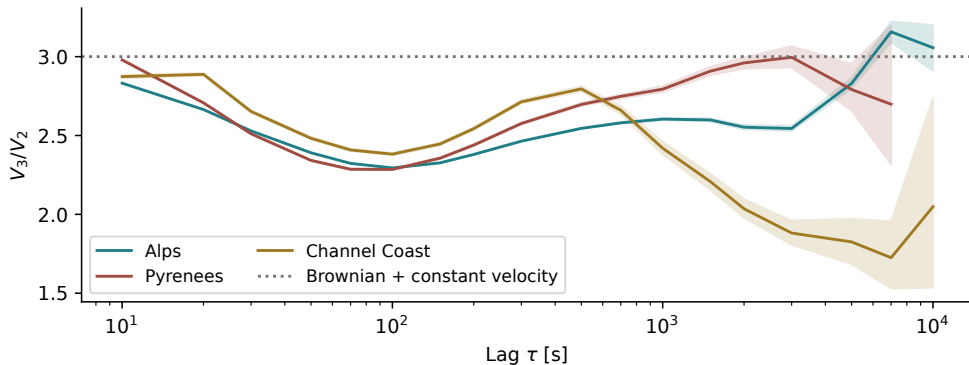


Common origins through 1000 s. Dashed: 23 shared context strata, descriptive. Shading: simultaneous 95% bands for the unstandardised ratios.

Thesis §3.5.2

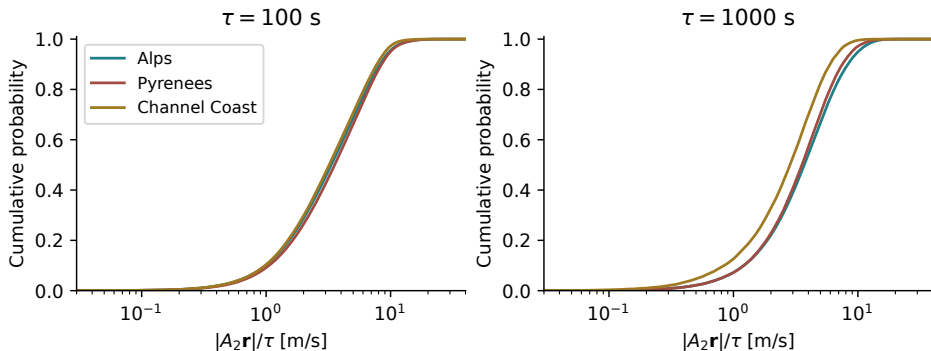
Third differences and the distribution of changes

The ratio between orders changes with the lag



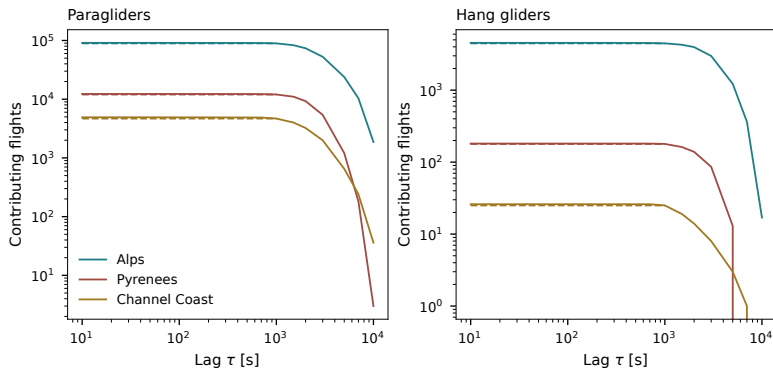
Matched origins and flight weights; pointwise 95% site-day intervals. Dotted: Brownian motion plus constant velocity.

The distribution distinguishes frequent small changes from large events



Equal-flight CDFs on matched-order support. Full report: all four display lags, zero and overflow mass, and quantile bin bounds.

Order-three support limits the long-lag regional comparison

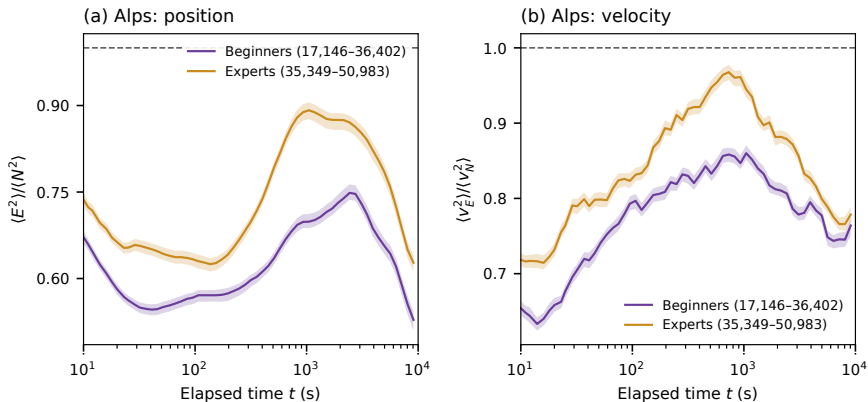


Solid: orders share 3τ support. Dashed: origins common through 1000 s. All eligible regional flights contribute where supported.

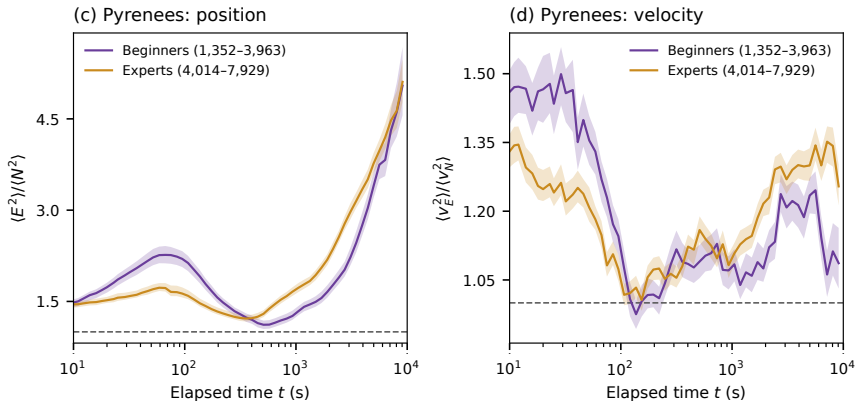
Thesis §3.5.3

Terrain, equipment and wind

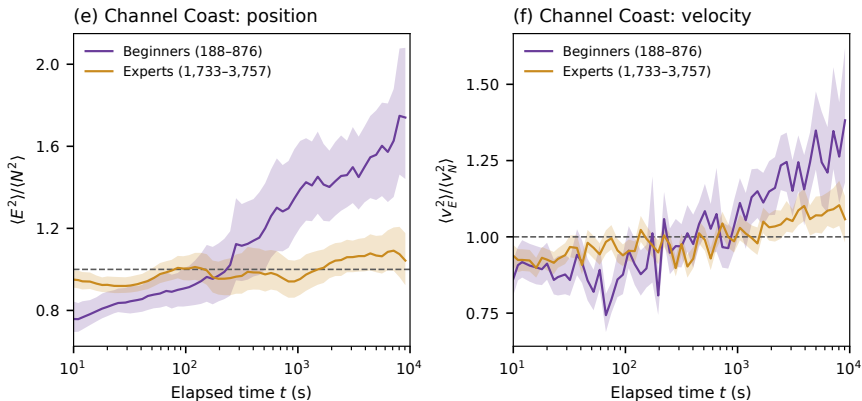
Alps: C/D/CCC has smaller coordinate-moment imbalance



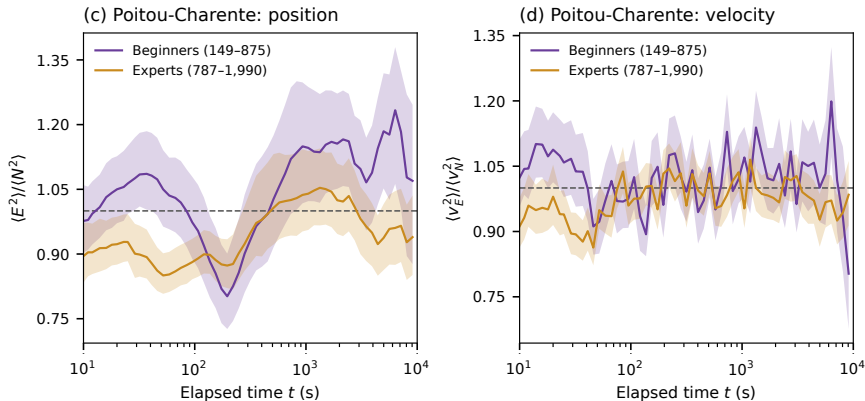
Pyrenees: the ordering of the two groups reverses



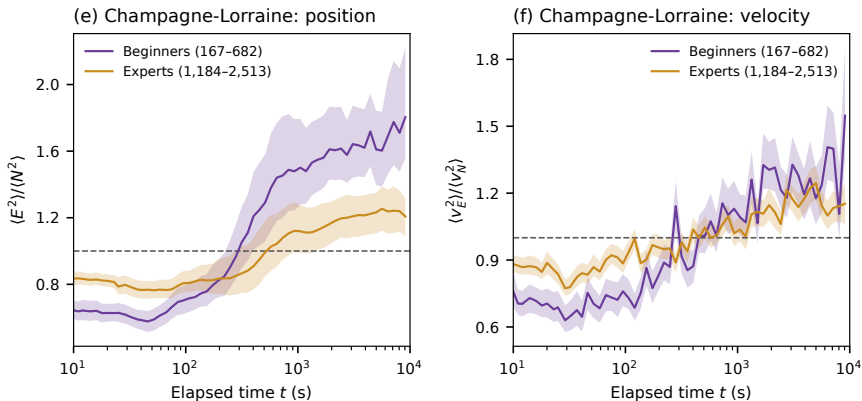
Channel Coast: the late position contrast is pronounced



Poitou-Charente: the group curves cross



Champagne-Lorraine: a late contrast does not keep C/D/CCC at unity



Hypothesis to discuss

Equipment and pilot choices may moderate directional constraints associated with wind and terrain. Similar mountain profiles may reflect shared route constraints.

Evidence limiting that interpretation

The Pyrenean ordering reverses; class is an experience proxy; coastal terrain is not absent; late support differs; and ensemble balance does not establish individual isotropy or performance.

Define the target before calling it resistance to wind

$$\mathbf{v}_{\text{ground}} = \mathbf{v}_{\text{air}} + \mathbf{w}_h.$$

Upwind progress, air-relative performance, choice of route and geographic moment balance are different outcomes.

A first comparison would match overlapping site-day, duration and task support; then separate mean drift, centred variability and individual route orientation. Regional references show terrain alignment and a remaining coastal wind-PCA offset at flight altitude; local track-matched observations are still needed.